

Title

Nigeria Financial Crime Intelligence & AML Risk Monitoring Platform

Subtitle

Portfolio Project | Power BI | DAX | AML Analytics

Prepared by

Victor Ikomi

Include:

- Tools used:
 - Power BI
 - Power Query
 - DAX
- Data Modelling
 - Dataset:

Kaggle NIBSS Fraud Dataset

Business Problem

Nigeria's financial system operates in a complex environment shaped by a large informal economy, identity-verification gaps, and the rapid expansion of NIBSS-driven (Nigeria Inter Bank Settlement System) digital payment channels. These conditions create persistent challenges for banks attempting to detect and mitigate financial-crime risks, particularly fraud typologies such as **SIM-swap-enabled account takeover, social-engineering-driven NIP transfers, and mule-account activity.**

Financial institutions are expected under **CBN AML/CFT guidelines** to maintain effective monitoring frameworks capable of identifying unusual patterns across **NIP, POS, USSD, mobile banking, and internet-banking channels.** This project demonstrates how transactional data can be transformed into **operationally meaningful insights** to support fraud detection, AML surveillance, and case prioritisation.

Objectives

This project set out to build a practical **Financial Crime Intelligence and AML Monitoring Platform** using Power BI and Nigerian transaction data. The key objectives were to:

- **Track fraud and AML risk** across major NIBSS channels (NIP, POS, USSD, Mobile, Web) using interactive dashboards.
- **Apply a risk-based scoring model** to classify transactions and highlight unusual behavior.
- **Identify high-risk customers, merchants, and locations** requiring enhanced monitoring or investigation.
- **Rank investigations** based on risk severity and alignment with known Nigerian fraud typologies.
- **Support AML decision-making** in line with CBN expectations for risk-based monitoring.
- **Convert raw NIBSS-style data into insights** that inform fraud and AML decision-making.

- **Demonstrate practical Power BI techniques**—DAX, Power Query, and star-schema modelling—for financial crime analytics in the Nigerian context.

Dashboard Pages

1. Nigerian Financial Crime Dashboard

Purpose: Provide a high-level view of **what happened**, highlighting major fraud trends, risk distribution, and crime

risk concentration.

2. Nigeria AML Risk Monitoring Dashboard

Purpose: Identify **where risk is concentrated**, with emphasis on NIP/USSD activity, geographic anomalies, risk trend over time and deviations from high-risk customer monitoring.

3. Nigerian Financial Crime Investigation Dashboard

Purpose: Prioritise **which cases require review**, surfacing high-risk transactions consistent with Nigerian fraud typologies and CBN-aligned AML red flags.

Dataset Description

The dataset used for this project is a Nigerian electronic-payments transaction file sourced from the NIBSS fraud dataset on Kaggle and it contains approximately one million transactions.

The core fields include transaction and customer identities, transaction amount, timestamp, channel information (e.g., POS, mobile, online), Location fields are available but not consistently populated. Merchant category, age group, fraud indicator. The fraud indicator is binary, with a small proportion of transactions marked as confirmed fraud and merchant composite risk score.

Additional engineered variables such as transaction velocity, frequency counts, and simple behavioral risk scores were created during preprocessing to mimic features used in production fraud-monitoring systems. These derived fields help highlight unusual

spending patterns or rapid transaction bursts that may indicate account takeover or mule activity.

Overall, the dataset provides a workable foundation for demonstrating fraud-detection concepts and AML-style risk analysis, but it does not capture the full operational complexity of Nigerian financial-crime typologies.

Data Model

Star Schema Design

The data was structured using a star schema model in Power BI. The main transactional table was designed as the fact table, while supporting dimension tables were created to improve filtering, reporting, and analytical performance.

Fact and Dimension Table

Fact_Transactions

- Transaction_id
- Customer_id
- Timestamp
- Amount
- Channel
- Location
- Merchant_category
- Is_fraud
- Fraud_technique
- Velocity_score
- Merchant_composite_risk
- Overall Risk Score
- Risk Band

Dimension Tables

- **Dim_Customer:** customer_id, age_group
- **Dim_Channel:** channel

- **Dim_Location:** location / map location
- **Dim_Merchant_Category:** merchant_category
- **Dim_Date:** date, month, month name, quarter, year, day of week, weekend flag

Each dimension table was linked to the Fact_Transactions table using one-to-many relationships, with the dimension tables on the “one” side and the transaction table on the “many” side.

Calculated Columns and Measures

In addition to the core data model, several calculated columns and DAX measures were developed to support fraud analysis, AML risk monitoring, and investigation prioritization.

Calculated Columns

- Date hierarchy attributes (Year, Quarter, Month Number, Month Name, Day of Week, Weekend Flag)
- Map Location
- Amount Risk Score
- Channel Risk Score
- Geographic Risk Score
- Frequency Risk Score
- Overall Risk Score
- Risk Band
- Merchant Risk Band
- Investigation Priority

DAX Measures

- Total Transactions
- Total Transaction Value
- Fraud Cases
- Fraud Rate (%)
- High-Risk Transactions
- High-Risk Customers
- High-Risk Merchants

- High-Risk Transaction Value
- Average Risk Score
- High-Risk Locations

These calculations enabled the development of risk-based analytics, performance indicators, geographic risk visualizations, customer monitoring views, and investigation dashboards.

Risk Scoring Methodology

A rule-based AML risk scoring model was developed to classify transactions by financial crime risk exposure. The model combines multiple risk factors commonly used in fraud and AML monitoring.

The overall risk score was calculated using four components:

Overall Risk Score = Amount Risk Score + Channel Risk Score + Geographic Risk Score + Frequency Risk Score

Risk Components

1. Amount Risk Score

Higher-value transactions are treated as higher-risk, especially when they exceed typical patterns for the customer's **BVN-linked profile (Bank Verification Number)**. Nigerian banks often apply enhanced scrutiny to large NIP transfers and high-value POS reversals due to known fraud typologies such as **account takeover**, **SIM swap fraud**, and **mule account activity**.

2. Channel Risk Score

Different **NIBSS channels** carry different levels of inherent risk:

- **NIP(NIBSS Instant Payments)** high-risk due to speed and irreversibility
- **USSD** often exploited in SIM-swap and social-engineering fraud
- **POS (Point-of-Sale)** associated with merchant-level fraud and terminal compromise, *routed through NIBSS*
- **Mobile/Web** elevated risk for account takeover and phishing-related activity
- **ATM/Branch** generally lower risk but still monitored for structuring
- **Branch / Teller** *in-person transactions*

The scoring reflects these relative exposures.

3. Geographic Risk Score

Geographic scoring considers the transaction's originating or beneficiary location. In practice, Nigerian banks reference:

- **High-risk states or regions** with elevated fraud incidence
- **Cross-border corridors** subject to CBN (Central Bank of Nigeria) AML scrutiny
- **Locations inconsistent with the customer's BVN history**

This helps identify red flags such as mule networks operating across state lines or sudden activity in unfamiliar regions.

4. Frequency Risk Score

Rapid transaction bursts within a short window (e.g, multiple NIP transfers within 10–30 minutes) are common indicators of:

- **Account takeover**
- **Mule account cash-outs**
- **Structuring to avoid reporting thresholds**

The score increases when a customer's 24-hour activity deviates from their BVN-linked behavioral baseline.

Risk Bands

High - Transactions requiring immediate review, typically involving high-risk channels (NIP/USSD), unusual locations, or patterns consistent with known fraud typologies.

Medium-High - Elevated-risk activity that may require follow-up depending on customer history and BVN profile.

Medium - Routine transactions with some risk indicators but not enough to justify escalation.

Medium-Low - Lower-priority items that align with expected customer behavior.

Purpose of Methodology

This scoring model provides a practical way to:

- Highlight transactions with elevated financial-crime risk
- Support AML monitoring aligned with **CBN expectations**
- Identify activity consistent with common Nigerian fraud typologies

- Help investigators prioritise cases for review

It is not intended to replace machine-learning models or enterprise-grade AML systems. Instead, it demonstrates how simple rule-based scoring can be applied to a NIBSS-style dataset in a way that reflects real operational practices.

Key Findings

Fraud Activity Concentration

Fraud incidents were not evenly distributed across the dataset. Most confirmed cases were linked to a small set of high-exposure NIBSS channels, particularly **NIP**, **USSD**, and **mobile banking**. These channels continue to be the primary vectors for **account takeover**, **SIM-swap-enabled fraud**, and **social-engineering-driven transfers**, consistent with industry trends reported across Nigerian banks.

Geographic Risk Exposure

High-risk activity was concentrated in a limited number of states that record elevated fraud attempts. Several transactions originated from or were routed to locations that did **not align with the customer's BVN-linked behavioural footprint**, a common red flag for mule accounts or compromised profiles. This pattern aligns with CBN expectations around monitoring unusual geographic deviations.

Risk Distribution

The majority of transactions fell into the Medium or Medium-High risk bands, which is typical for a retail-payments environment with high digital-channel usage. Only a small proportion of transactions were classified as High risk, but these cases exhibited multiple red flags — such as rapid NIP transfers, unusual locations, warranting immediate review.

High-Risk Transaction Identification

The rule-based scoring model effectively reduced the dataset of over one million transactions to a manageable set of critical-risk alerts. These alerts reflected patterns commonly escalated in Nigerian AML and fraud-monitoring

teams, such as rapid outbound transfers, sudden high-value NIP activity, and transactions inconsistent with historical BVN activity.

Customer Risk Concentration

High-risk activity was concentrated among a relatively small number of BVN-linked customer profiles. This mirrors real-world patterns where a limited set of accounts — often **mule accounts**, **compromised profiles**, or **synthetic identities** — account for a disproportionate share of suspicious activity. This concentration supports targeted enhanced due-diligence and customer-level monitoring.

Fraud Techniques

The confirmed fraud cases were dominated by social engineering, including **phishing**, **vishing**, and **SIM-swap-enabled account takeover**. These typologies remain the most prevalent across Nigerian banks and highlight the need for stronger behavioral analytics, customer education, and SIM-swap monitoring controls.

Channel Risk

Digital channels — particularly internet banking, mobile banking, and USSD — recorded higher average risk scores than traditional channels such as **ATM** or **branch**. This reflects the broader shift toward digital payments within the NIBSS ecosystem and the corresponding increase in fraud attempts targeting these channels.

Recommendations

Strengthen Monitoring of High-Risk Transactions

Transactions flagged as **High risk** especially those involving **rapid NIP transfers**, **USSD activity following SIM-swap events**, or **behavior inconsistent with the customer's BVN profile** should be escalated for immediate review. Timely intervention is critical given the speed and irreversibility of NIBSS instant payments.

Enhance Channel-Based Controls

Digital channels such as **NIP**, **USSD**, **mobile banking**, and **internet banking** continue to carry the highest fraud

exposure. Banks should strengthen authentication, behavioral analytics, and anomaly-detection controls on these channels, in line with **CBN's risk-based cybersecurity and AML expectations**.

Adopt Risk-Based Investigation Prioritisation

Investigation teams should prioritise customers, merchant categories, and locations showing **clustered high-risk activity**, such as repeated NIP cash-outs, unusual geographic patterns, or transactions routed through known mule-account hotspots. This improves investigative efficiency and reduces false positives.

Increase Focus on Fraud Typologies

Fraud patterns in Nigeria remain dominated by **social engineering**, **SIM-swap-enabled account takeover**, and **mule-account operations**. Institutions should strengthen customer-awareness programs, enhance SIM-swap monitoring, and continuously update typology libraries to reflect emerging threats across NIBSS channels.

Implement Dynamic Risk Scoring

Risk-scoring models should be realigned regularly to reflect evolving transaction patterns, seasonal fraud spikes, and changes in customer behavior. Incorporating **BVN-linked behavioral baselines** and **channel-specific risk weights** will improve detection accuracy.

Leverage Data Analytics for AML Monitoring

Advanced analytics, machine-learning-assisted scoring, and interactive dashboards should be integrated into AML and fraud-monitoring workflows. These tools help analysts quickly identify anomalies across **NIP, POS, USSD, mobile, and web** channels and support more informed decision-making.

Maintain Ongoing Risk Assessments

Regular reviews of customer, channel, geographic, and merchant-category risks are essential to ensure controls remain aligned with the institution's risk profile. This includes periodic reassessment of **high-risk states**, **BVN-linked behavioral deviations**, and **merchant categories with elevated fraud exposure**.

Data Source Limitations

Use of a Public, Synthetic Dataset

The analysis is based on a publicly available dataset that simulates NIBSS-style transactions. While useful for demonstration, it does **not fully capture the operational**

complexity of real Nigerian banking data, including institution-specific controls, customer-level behavior, or internal fraud-monitoring metadata.

Limited Customer Information

The dataset contains only basic customer identifiers and lacks deeper **BVN-linked demographic and behavioral attributes**. In practice, Nigerian banks rely heavily on BVN history, device intelligence, account age, and behavioral baselines to assess customer risk. The absence of these fields limits the ability to perform **comprehensive customer-risk profiling**.

Absence of AML-Specific Indicators

Key AML attributes required under **CBN AML/CFT guidelines** such as customer occupation, source of funds, PEP status, sanctions exposure, and beneficial-ownership information are not included. Without these fields, the

dataset cannot fully support **risk-based AML monitoring** or customer-due-diligence assessments.

Limited Geographic Coverage

The dataset includes transactions from only a subset of Nigerian states. Real-world fraud and AML monitoring requires visibility across all regions, especially high-risk corridors and states with elevated fraud incidence. This limitation reduces the ability to analyse **geographic risk concentration** accurately.

Assumptions in the Risk-Scoring Model

The risk-scoring framework relies on simplified rules for transaction amount, channel, geography, and frequency. In a real NIBSS-connected environment, risk scoring would incorporate:

- BVN behavioral baselines
- Device and SIM-swap intelligence
- Channel-specific fraud patterns (e.g., NIP cash-outs, USSD compromise)
- Merchant-category risk ratings
- Historical fraud losses

Because these elements are not available, the model reflects a **conceptual demonstration** rather than a production-grade scoring engine.

Conclusion

This project demonstrates how **NIBSS-style transaction data** can be transformed into actionable intelligence to strengthen **fraud prevention** and **AML risk monitoring** across Nigerian digital-payment channels. Using Power BI, the solution provides a unified view of financial-crime exposure across **NIP, USSD, POS, mobile, and internet-banking channels**, enabling faster identification of high-risk customers, suspicious transactions, and geographic hotspots.

The risk-scoring approach, built around **amount, channel, geographic deviation, and transaction frequency** offers a practical framework for prioritising alerts, especially for patterns linked to **NIP cash-outs, SIM-swap-related USSD activity, and behavior inconsistent with BVN history**.

Although the dataset is publicly available and lacks full AML attributes, the project clearly shows the value of combining **risk scoring, behavioral analysis, and interactive dashboards** to support **CBN-aligned, risk-based monitoring**.

Future enhancements such as **customer risk ratings, sanctions/PEP screening, network-link analysis, and machine-learning models** would further strengthen the platform and bring it closer to a production-grade

financial-crime monitoring environment.